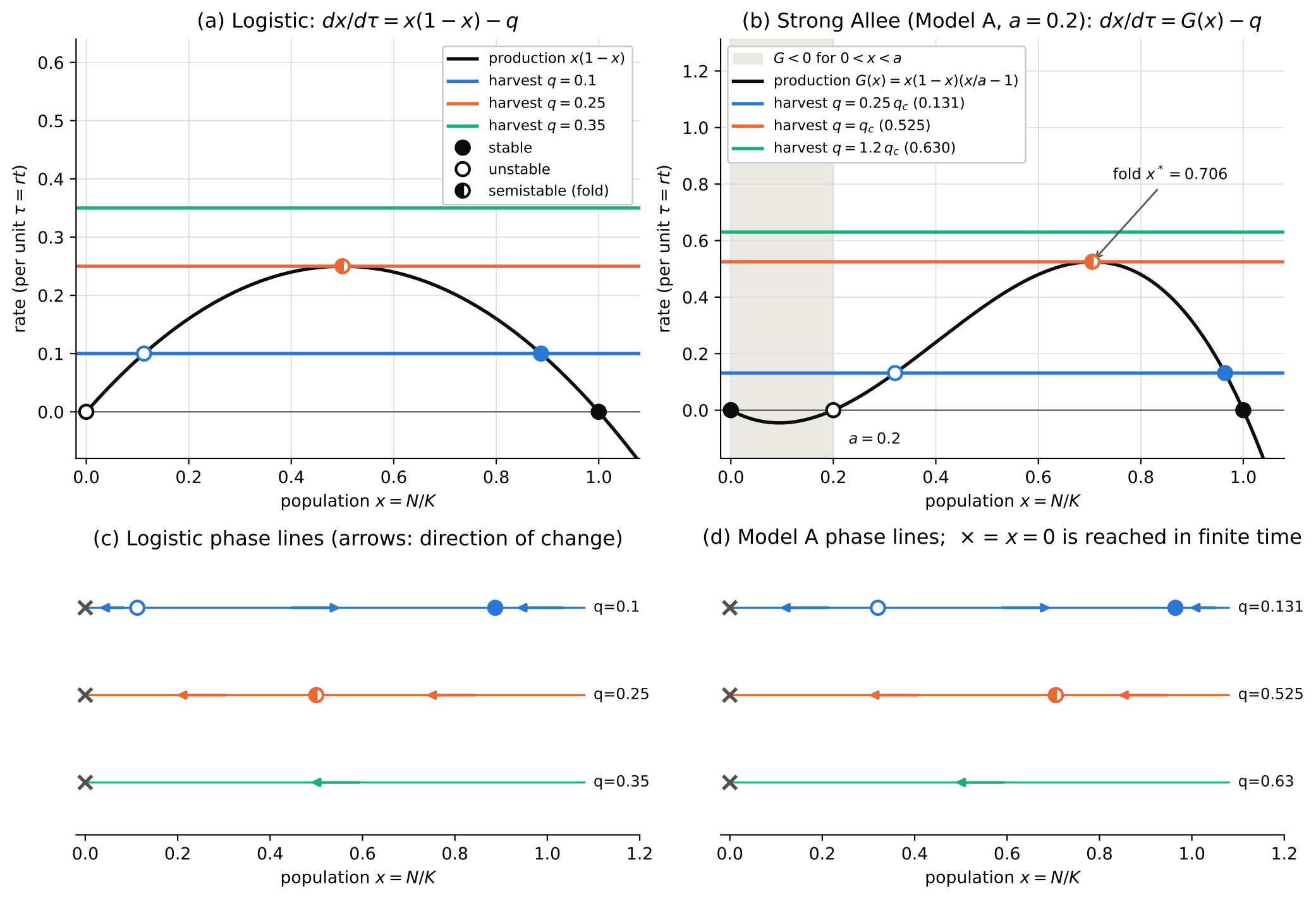
(a) Logistic: $dx/d\tau = x(1-x) - q$
production $x(1-x)$
harvest $q = 0.1$
harvest $q = 0.25$
harvest $q = 0.35$
stable
unstable
semistable (fold)
rate (per unit $\tau = rt$)
population $x = N/K$
(b) Strong Allee (Model A, $a = 0.2$): $dx/d\tau = G(x) - q$
$G < 0$ for $0 < x < a$
production $G(x) = x(1-x)(x/a - 1)$
harvest $q = 0.25\,q_c$ (0.131)
harvest $q = q_c$ (0.525)
harvest $q = 1.2\,q_c$ (0.630)
fold $x^* = 0.706$
$a = 0.2$
rate (per unit $\tau = rt$)
population $x = N/K$
(c) Logistic phase lines (arrows: direction of change)
q=0.1
q=0.25
q=0.35
population $x = N/K$
(d) Model A phase lines; × = $x = 0$ is reached in finite time
q=0.131
q=0.525
q=0.63
population $x = N/K$